

ASSIGNMENT NO .3

Problem 1: Sum of Two Numbers (Using a Method)

Problem Statement:

Write a Java program that includes a method to calculate the sum of two numbers.

1. Create a method `sumOfTwoNumbers()` that takes two integers as parameters, calculates their sum, and returns the result.
2. In the main method, use the `Scanner` class to prompt the user to enter two integers.
3. Pass the user inputs to the `sumOfTwoNumbers()` method and print the sum.

Code:

```
import java.util.Scanner;
public class SumOfTwoNumbers {
    public static int sumOfTwoNumbers(int a, int b) {
        return a + b;
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int num1 = scanner.nextInt();

        System.out.print("Enter second number: ");
        int num2 = scanner.nextInt();

        int sum = sumOfTwoNumbers(num1, num2);
        System.out.println("The sum of " + num1 + " and " + num2 + " is " + sum + ".");
    }
}
```

Expected Output:

```
Enter first number: 15
Enter second number: 25
The sum of 15 and 25 is 40.
```

Problem 2: Simple Age Checker (Using a Method)

Problem Statement:

Write a Java program that includes a method to check the age category.

1. Create a method `checkAgeCategory()` that takes an integer (age) as a parameter and prints whether the user is a minor, adult, or senior citizen.
2. In the main method, use the Scanner class to prompt the user to enter their age.
3. Pass the user's age to the `checkAgeCategory()` method.

Code:

```
import java.util.Scanner;

public class AgeChecker {
    public static void checkAgeCategory(int age) {
        if (age < 18) {
            System.out.println("You are a minor.");
        } else if (age < 60) {
            System.out.println("You are an adult.");
        } else {
            System.out.println("You are a senior citizen.");
        }
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the age: ");
        int age = scanner.nextInt();
        checkAgeCategory(age);
    }
}
```

Expected Output:

Enter the age: 30
You are an adult.

Problem 3: Print Even Numbers (Using while Loop)

Problem Statement:

Write a Java program that prints all even numbers between 1 and 50 using a while loop.

1. Create a method printEvenNumbers() that prints all even numbers from 1 to 50.
2. Use a while loop to iterate from 1 to 50 and print the even numbers.

Code:

```
public class EvenNumbers {  
  
    public static void printEvenNumbers() {  
        int num = 1;  
        while (num <= 50) {  
            if (num % 2 == 0) {  
                System.out.print(num + " ");  
            }  
            num++;  
        }  
        System.out.println();  
    }  
  
    public static void main(String[] args) {  
        printEvenNumbers();  
    }  
}
```

Expected Output:

2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50

Problem 4: User Input for Positive Numbers (Using do-while Loop)

Problem Statement:

Write a Java program that repeatedly asks the user to enter a positive number.

1. Create a method `askForPositiveNumber()` that uses a do-while loop to ask the user for a number until they enter a positive number.
2. Use the `Scanner` class to take the user's input.
3. Once a positive number is entered, the program should display the number.

Code:

```
import java.util.Scanner;

public class x    PositiveNumber {
    public static int askForPositiveNumber() {
        Scanner scanner = new Scanner(System.in);
        int number;

        do {
            System.out.print("Enter a positive number: ");
            number = scanner.nextInt();
        } while (number <= 0);
        return number;
    }

    public static void main(String[] args) {
        int ans = askForPositiveNumber();
        System.out.println("You entered a positive number: " + ans);
    }
}
```

Expected Output:

```
Enter a positive number: -5
Enter a positive number: 0
Enter a positive number: 8
You entered a positive number: 8
```

Problem 5: Print Multiplication Table (Using for Loop)

Problem Statement:

Write a Java program that prints the multiplication table for a given number (e.g., number 5) using a for loop. The program should:

1. Create a method printMultiplicationTable() that takes a number as a parameter and prints its multiplication table from 1 to 10.
2. Use a for loop to iterate through numbers 1 to 10 and print the multiplication results.

Code:

```
import java.util.Scanner;

public class MultiplicationTable {
    public static void printMultiplicationTable(int number) {
        for (int i = 1; i <= 10; i++) {
            System.out.println(number + " x " + i + " = " + (number * i));
        }
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int number = scanner.nextInt();
        printMultiplicationTable(number);
    }
}
```

Expected Output:

Enter a number: 5

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

Problem 6: Calculate the Sum of Numbers from 1 to N (Using for Loop)

Problem Statement:

Write a Java program that calculates the sum of all integers from 1 to N (where N is a positive integer) using a for loop. The program should:

1. Create a method calculateSum() that takes a number N and calculates the sum of all integers from 1 to N.
2. Use a for loop to iterate through all integers from 1 to N and add them up.

Code:

```
import java.util.Scanner;
public class SumOneToN {
    public static int calculateSum(int n) {
        int sum = 0;
        for (int i = 1; i <= n; i++) {
            sum += i;
        }
        return sum;
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a positive integer Number: ");
        int n = scanner.nextInt();

        int result = calculateSum(n);

        System.out.println("The sum of all integers from 1 to " + n + " is " + result + ".");
    }
}
```

Expected Output:

Enter a positive integer Number: 5
The sum of all integers from 1 to 5 is 15.

Bonus Problem: Menu-Driven Java Program (Switch-Case)

Problem Statement:

You are required to write a menu-driven Java program that implements four separate problems. The program should allow the user to select which problem to run, execute the corresponding logic, and then return to the menu until the user chooses to exit.

The four problems are as follows (Already done in assignment 2, just put it in switch case):

Problem 1: Grade Evaluation System

Problem 2: Leap Year Check

Problem 3: Day of the week

Problem 4: Identify Default Values of Variables

Case 5: Exit

Code:

```
import java.util.Scanner;

public class MenuDrivenProgram {
    static byte a;
    static short b;
    static int c;
    static long d;
    static float e;
    static double f;
    static char g;
    static boolean h;

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;

        do {
            System.out.println("1. Grade Evaluation System");
            System.out.println("2. Leap Year Check");
            System.out.println("3. Day of the Week");
            System.out.println("4. Identify Default Values of Variables");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
```

```

switch (choice) {
    case 1: {
        int maths = 80;
        int science = 85;
        int history = 90;

        int average = (maths + science + history) / 3;
        String grade;

        if (average >= 90) {
            grade = "A";
        } else if (average >= 70) {
            grade = "B";
        } else if (average >= 50) {
            grade = "C";
        } else if (average >= 30) {
            grade = "D";
        } else {
            grade = "Fail";
        }

        System.out.println("Average Marks: " + average);
        System.out.println("Grade: " + grade);
        break;
    }

    case 2: {
        System.out.print("Enter a year: ");
        int year = sc.nextInt();

        boolean isLeap = (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);

        if (isLeap) {
            System.out.println(year + " is a leap year.");
        } else {
            System.out.println(year + " is not a leap year.");
        }
        break;
    }
}

```



```

case 3: {
    System.out.print("Enter a day number (1-7): ");
    int dayNumber = sc.nextInt();

    switch (dayNumber) {
        case 1:
            System.out.println("The day is Monday");
            break;
        case 2:
            System.out.println("The day is Tuesday");
            break;
        case 3:
            System.out.println("The day is Wednesday");
            break;
        case 4:
            System.out.println("The day is Thursday");
            break;
        case 5:
            System.out.println("The day is Friday");
            break;
        case 6:
            System.out.println("The day is Saturday");
            break;
        case 7:
            System.out.println("The day is Sunday");
            break;
        default:
            System.out.println("Invalid day number");
    }
    break;
}

case 4: {
    a = 10;
    b = 200;
    c = 100000;
    d = 1000000000000L;
    e = 3.14f;
    f = 3.14159;
}

```

```

        g = 'A';
        h = true;
        System.out.println("\n--- After initialization ---");
        System.out.println("byte a  = " + a);
        System.out.println("short b  = " + b);
        System.out.println("int c    = " + c);

        System.out.println("long d   = " + d);
        System.out.println("float e  = " + e);
        System.out.println("double f = " + f);
        System.out.println("char g   = " + g);
        System.out.println("boolean h = " + h);
        break;
    }
    case 5:
        System.out.println("Exit");
        break;

    default:
        System.out.println("Invalid choice. Please select between 1 and 5.");
    }
} while (choice != 5);
}

```

Output:

1. Grade Evaluation System
2. Leap Year Check
3. Day of the Week
4. Identify Default Values of Variables
5. Exit

Enter your choice: 1
Average Marks: 85
Grade: B

1. Grade Evaluation System
2. Leap Year Check
3. Day of the Week
4. Identify Default Values of Variables
5. Exit

Enter your choice: 5
Exit